

Pricing Public Housing on a Moving Market: An End-to-End Study of Singapore HDB Resale Transactions, 2017–2026

Abstract

Singapore's HDB resale market is not a set of independent house prices. It is a thick, regulated, high-frequency auction in which location, remaining lease, dwelling type, and the official policy environment jointly determine what buyers pay. This project reconstructs that market from registration-level transactions (2017 onward), joins planning-area and transport geography without leaking future information into every model, and asks a harder question than in-sample fit: Can a model trained only through 2025 price the 2026 book?

The work is organised as evidence, not as a software tour. First, the transaction tape is cleaned to a single grain and read with SQL and exploratory plots: volume, median price, and price per square metre do not always move together, because mix (town, flat type, size) changes. Second, distance to the city centre and—separately—distance to MRT are treated as distinct economic objects; MRT is withheld from the conservative model because stations appear over the sample. Third, two estimators are trained on a temporal split. Version 1 refuses MRT and raw coordinates. Version 2 uses the full geocode cache and is the engine behind an indicative calculator. Neither model uses ``price_per_sqm``, which is a transformation of the target.

The claim of the article is modest and operational: A 2026 holdout, a median baseline, residual cuts by town, type and price band, and a user-facing number published with its test MAE and a valuation disclaimer, are more honest than a leaderboard R^2 on a shuffled sample.

Introduction

Public housing in Singapore is unusual among large cities: The majority of residents live in HDB flats, a large secondary market exists, and prices are observed as registered transactions, not asking prices. That makes the tape unusually complete—and unusually tempting to overfit.

Three mistakes dominate amateur HDB models.

1. Leakage. ``price_per_sqm`` is resale price divided by floor area. Feeding it to model that's asked to predict price isn't feature engineer. It's restating the model.

2. Random splits. 2026 prices already embed 2026 rates, cooling measures, and supply. Mixing years in train and test measures interpolation, not forecasting.
3. Timeless maps. A walk-time to a station that opened in 2022 is not a 2018 attribute. Using today's MRT layer on a 2017 sale inflates the apparent value of "being near MRT" for years when that station did not exist.

This project is built against those three mistakes. The empirical objects are: The cleaned transaction table; Official context series (resale price index, applications, published medians) that must not be treated as the same grain as a completed deal; A geocode cache produced once and then frozen; Two model generations; And two reading surfaces—a five-page market dashboard and a calculator that only predicts after the address is found in the cache.

What follows is an analysis of that system: What the market looks like, which associations are structural, how the models are identified, and where they should not be believed.

Data and measurement

The transaction as the unit of analysis

One row is one registered resale. The economic variables are time of registration, town, flat type and model, floor area, storey, remaining lease, and consideration (resale price). Storey arrives as a band (07 TO 09) and is expanded to lower, midpoint and upper floors so that a tree can use a number and a form can still speak in HDB language.

Remaining lease is stored in months. A slider in "years" in the calculator is a display convention; the model sees ``remaining_lease_month``. Using both ``flat_year`` and remaining lease in the same specification is close to double-counting age. Version 2 keeps one of the pair.

Cleaning that changes the science

Missingness in HDB auxiliary files is often encoded as ``NA`` or ``-``, not as blank. Leaving those as categories produces fake "towns". Category strings are standardised (3-room → 3 ROOM, Ang Mo Kio → ANG MO KIO) before any join; Otherwise, the join is a silent sample filter, which is a form of selection bias.

Geography, frozen

Addresses are (block, street). The geocode cache maps that pair to latitude, longitude, planning-area identity (zone_id, subzone), nearest MRT (from LTA station-exit geography after the live "nearest MRT" API proved unreliable), distance to that station, and distance to Raffles Place as a city-centre proxy.

Two modelling consequences follow.

1. zone_id is a name, not a number. Encoding it as a continuous feature invents an order that URA did not define.
2. Distances are properties of an address at cache time. Version 1 therefore omits MRT entirely. Version 2 uses MRT because the calculator prices flat today, on today's map, and states that limitation in the model card.

OneMap tokens expire in days. After the cache exists, the estimator is offline. That is a measurement decision as much as an engineering one: The feature must be reproducible.

What the tape says about the market

The SQL layer and the first dashboard page are designed around a single question: What is happening? Volume, median price, median price per square metre, and median size are reported together because any one of them alone misleads.

Consider two expansions. If more large Executive flats trade, median price rises even if every size slice is flat. If the mix shifts to smaller units in expensive towns, median PPS can rise while median price falls. The dual-axis monthly chart (volume as columns, median price as a line) is there to stop the reader from narrating a “boom” from price alone in a thin month.

A year-on-year median is defined on the latest selected year versus the year before it. Averaging YoY across a 2017–2026 slicer is not a market fact; It is an artefact of the filter.

Location is read at subzone, not at the transaction point. A map of 200,000 dots is not more precise; It is a heatmap of density that the eye cannot audit. Size of the bubble is count. Beside the map, top and bottom subzones by PPS, and PPS by MRT-distance and city-distance bands, ask whether proximity is priced. The project does not claim that a 400 m walk “causes” a premium. Towns near MRT are also older, more central, or more constrained. The charts are associations with sample size in the tooltip.

Property drivers repeat that discipline. Floor area is plotted against median price (a larger flat costs more dollars). Storey and remaining lease are plotted against median PPS (the market's unit price). A town × flat-type matrix of PPS is the simplest interaction the data support: A four-room is not a homogeneous good.

Official statistics are a context, not a second target. Transaction medians can drift from the Resale Price Index because the index is quality-adjusted in ways a raw median is not. Completions can lag applications. Published medians for registered applications are not the same sample as completed deals. The sixth dashboard page exists so those discrepancies are visible rather than “corrected” in the model.

Identification of the price model

The estimate

The object is $\mathbb{E}[P \mid X]$ for a completed registration, with X restricted to attributes known at listing: size, type, storey, remaining lease, time, and (depending on version) location. It is not a hedonic causal effect of building an MRT station, and not a valuation under the HDB or professional-appraisal rules.

The split

Let t be registration year.

$$\mathcal{D}_{\text{train}} = \{i: 2017 \leq t_i \leq 2025\}, \mathcal{D}_{\text{test}} = \{i: t_i = 2026\}.$$

Every metric quoted for model choice is computed on $\mathcal{D}_{\text{test}}$ only. That is a single future year. It is a stricter test than a random 20% of 2017–2026, and a weaker test than a sequence of rolling one-year-ahead exercises. The article treats 2026 as one stress year, not as “the” generalisation of the model.

What is excluded

``price_per_sqm`` is excluded by construction. ``search_val`` is a geocoding query string. status is constant. Version 1 also drops any column whose name encodes MRT. Version 2 drops only the redundant age/lease twin.

The preprocessor is part of the model. Numeric fields are median-imputed and scaled; Categoricals are imputed and one-hot encoded with unknown levels ignored and a cap on cardinality. The object saved to disk is the pipeline, so a calculator cannot apply a new one-hot map to an old set of coefficients.

Why a median baseline

HDB prices are right skewed. A mean dummy is an easy opponent. The global training median is the right naïve forecast for a MAE-oriented product. Boosted trees are interesting only to the extent they beat that median in 2026, not to the extent they fit 2019.

Linear and ridge models are retained as instruments. If they approach the trees, most of the predictable variation is additive (size, type, year). If they lag far behind, the market is interaction-heavy (town $\times$ type $\times$ lease). That comparison is the analysis; The zoo is not a beauty contest.

Difference between version I and version II

Version 1 asks: *How much of 2026 prices is already in dwelling attributes, calendar time, planning zone, and distance to the city, without using a 2026 rail map?*

Version 2 asks: *For a tool that prices a flat this week, how much does the current MRT layer add?*

Publishing only v2 would hide the time-inconsistency of transport. Publishing only v1 would give the calculator a worse address engine. The project keeps both.

How to read the holdout

The article's method does not require inventing the numbers; It requires not lying about what they mean.

Estimator	RMSE	MAE	MAPE	R^2
Training-set median	272241.6	204543.5	27.38727	-0.61835
Linear regression	78166.78	54120.38	8.252102	0.866584
Ridge	78161.65	54114.65	8.25108	0.866602
Random forest	41660.67	28693.42	4.35777	0.962102
XGBoost	38710.9	27236.59	4.165744	0.967279
LGBM	42895.91	30547.13	4.664947	0.959821

RMSE is dominated by the right tail—central, large, or unusual flats. A model that wins RMSE may still be a poor calculator for a typical four-room if it sacrifices the centre of the distribution.

MAE is the band the calculator prints. It is an average absolute error in 2026, not a 90% interval, and not a guarantee for 2027.

R^2 on a future year measures how much 2026 cross-sectional variation around the 2026 mean is captured. It can be high while the whole market has shifted; It can be modest while ranks within towns are accurate. It is not “the model explains Singapore housing.”

The residual plots are the results section that matter more than the leaderboard.

- Actual versus predicted. A bow above the diagonal at high prices means the model under-prices the tail—the region where MAE-as-range is most misleading.
- Residual histogram. A mean residual away from zero is **bias** in 2026, not noise. Bias is a market-level miss (the 2026 intercept moved). Noise is idiosyncratic.
- MAE by town. A calculator that is good in Tampines and poor in the Core Central Region is not “80k MAE”; It is two products.
- MAE by flat type. Executive and 2-room samples are thin; Errors there are not evidence about 4-room mass.

- MAE by actual price band. If error rises with price, a single $\pm$ MAE is too wide downstairs and too narrow upstairs. That fact should travel into the UI, not only into an appendix.

Comparables on the calculator—recent sales on the same block, then street, then subzone, with similar area—are a non-parametric check. If the model says \$650k and the last six similar sales sit at \$780–820k, the point estimate has lost the local level. The user should see both.

From estimate to claim

The Streamlit form is the identification made visible. Street and block are not decorative: They are the key into the cache. If the pair is absent, the project refuses to predict. Imputing a town-level centroid would look complete and be false.

Time is month-grain because the model was trained on ``trx_year`` and ``trx_month``. A user who wants “the value next year” is asking for a scenario, not a measurement; The year control is the scenario.

The disclaimer is part of the result, not legal decoration:

Estimates are indicative market-price predictions based on historical transactions, not official valuations.

HDB, banks, and valuers use other information (condition, restrictions, financing). This model has none of that. Publishing a sharp number without MAE, town-level error, and the disclaimer would complete the pipeline and destroy the analysis.

Limitations

The 2026 test is one year. Cooling measures, rate paths, and BTO completion waves can move the intercept faster than any hedonic residual. A responsible sequel is a rolling origin: train through T , test $T + 1$, for several T .

The MRT layer is a photograph. Version 2 should not be used to “value the opening of Thomson–East Coast Line in 2018.”

Condition, facing, stack, and seller urgency are omitted. Large residuals in mature estates are expected.

One-hot caps on block and postal collapse rare addresses into another bin. The premium for a specific stack is not in the model; It may appear in the comparables table, which is why that table is not optional colour.

Official series differ in definition and lag. They contextualise the tape; They are not extra columns in X unless rebuilt at transaction grain with care about leakage (future index values).

The calculator's support is the cache. New BTO blocks are out of sample until they are geocoded. That is a feature of the measurement design.

Conclusion

The contribution of the project is not a particular booster score. It is a consistent estimate: A 2026 price, from 2017–2025 registrations, without using a function of the price as a feature, with geography either frozen and time-unsafe (v2) or deliberately poorer and safer (v1), and with errors sliced where a buyer actually lives.

Read that way, the dashboard is the descriptive half of the same argument, the holdout table is the predictive half, and the calculator is the argument made expensive to misuse.